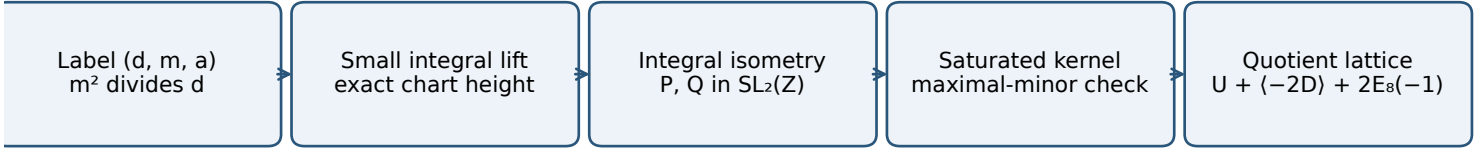

Exact integer certificates accompany every stage; the two $E_8(-1)$ factors are fixed.